

# FURKAN KARAMAN

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## SUMMARY

Chemistry student at METU with progressive laboratory experience from clinical analysis to geochemical testing, doping control, and analytical method development (SPME). Skilled in LC-MS/MS, GC-MS/MS, ICP-OES, XRF, HPLC, GC, AAS, and FTIR. Experienced in high-throughput quality control and trace analysis. Strong analytical mindset with a business management background. Highly motivated to learn daily, improve continuously, and contribute effectively in R&D or chemistry roles.

## EDUCATION

### **B.Sc. in Chemistry – Middle East Technical University (METU-ODTÜ), (2021-2025, Ankara/Türkiye)**

Developed strong theoretical knowledge, applied laboratory and reporting experience through 90 chemistry laboratory experiments covering Physical, Organic, Inorganic and Instrumental Analysis. Gained individual experience in spectroscopic techniques (IR, FTIR, UV-Vis, NMR), chromatographic methods (HPLC, GC, TLC) and atomic spectroscopy (AAS, AES); and has experience in qualitative and quantitative analysis, trace metal determination and instrumental data interpretation.

### **Associate Degree in Biochemistry – Karadeniz Technical University (2018-2020, Trabzon/Türkiye)**

They completed comprehensive theoretical and practical training in clinical, forensic, food, vitamin & hormone biochemistry and microbiology, including bacterial culture, antibiotic testing, biomolecule analysis, biochemical analysis, and routine laboratory equipment use.

## EXPERIENCE

### **Student Researcher – METU (ODTÜ), Prof. Dr. Ezel Boyacı Laboratory (08.2025-01.2026, Ankara/Türkiye)**

An innovative SPME-MOI-MS method was developed for the determination of L-Histidine for the early diagnosis of malignant hyperthermia; HLB, HLB-SAX, and PMAA extractive phases were synthesized and characterized, PAN-based fiber coatings were prepared, extraction and desorption conditions were optimized, MS injection parameters were adjusted, desorption solvent and time optimizations were performed, calibration curves were created, and the accuracy, sensitivity, and analytical performance of the method were evaluated for clinical applications by conducting recovery (%Recovery) and reproducibility (%RSD) studies.

### **Quality Control Intern—Hacettepe University Doping Control Center (07.2025-08.2025, Ankara/Türkiye)**

Performed advanced instrumental analyses using GC-MS/MS, LC-MS/MS, GC-C-IRMS, and LC-HR-MS for the detection and quantification of glucocorticoids, anabolic agents, stimulants, peptide hormones, narcotics, and steroids in urine samples. He strengthened his expertise in sample preparation, solid-phase extraction (SPE), MSTFA derivatization, internal standard calibration, quality control procedures, and high-precision trace-level analytical workflows, gaining practical experience in WADA-compliant and ISO/IEC 17025 compliant analytical workflows.

### **Analytical Chemistry Intern—Mineral Research and Exploration Institute (MTA) (07.2024-08.2024, Ankara/Türkiye)**

Accomplished high-throughput geochemical analyses of up to 240 soil samples per day using XRF, ICP-OES, and ICP-MS for the analysis of oxides and trace elements. He gained hands-on experience in sample preparation and quality control procedures using certified reference materials (CRM). He conducted thermogravimetric analysis (TGA), CHNS elemental analysis, bomb calorimetry, and density measurements on geological and coal samples to characterize mineralogical composition, fuel properties, and geochemical behavior for mining and energy exploration.

### **Biochemistry intern—Health Sciences University Ahi Evren Chest, Heart, and Vascular Surgery Training and Research Hospital (06.2019-07.2019, Trabzon/Türkiye)**

Conducted antibiotic susceptibility testing on more than 80 bacterial isolates to determine resistance patterns using disk diffusion and MIC methods. He cultured and identified pathogens from clinical samples using selective media. He performed biochemical, hemogram, and coagulation tests on more than 100 patients daily. He operated and calibrated biochemistry and hematology analyzers to ensure high diagnostic accuracy and efficiency.